

Question	Answer	Mark	Mark scheme	Additional guidance
23	73	P1	for process to work with cosine eg $(DB^2 =) 12^2 + 9^2 - 2 \times 12 \times 9 \times \cos 60$ (=117)	
		P1	for a correct order of operations to find DB, eg $\sqrt{225 - 216 \times \cos 60}$ or $\sqrt{117}$ (=10.8...)	$\sqrt{117} = 3\sqrt{13}$ Can be implied by correct use in subsequent calculations
		P1	for working with sine rule and 60° eg $\frac{\sin CBD}{9} = \frac{\sin 60}{[DB]}$ or $\sin CBD = 0.720...$ or $CBD = 46.10...$	3 rd and 4 th P1 can be awarded in any order [DB] must be clearly identified but may not be correct, but cannot be 6 or 9 or 12
		P1	for working with sine rule and 125° eg $\frac{\sin ABD}{6} = \frac{\sin 125}{[DB]}$ or $\sin ABD = 0.454...$ or $ABD = 27.02...$	CBD and ABD may be given as a variable, award marks unless contradicted
		A1	for answer within the range 72.5 to 73.5	